MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548016

(GENERATOR, RCFD)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the range control fine delay generator (RCFD generator) used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or requiest for porposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

F10548016	Assembly, RCFD Generator

PACKAGING DATA SHEET

10548016	Packaging of RCFD Generator

FSC 1285

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(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548016 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. - When supplied with the input signals and logic levels specified in Table I and Table II respectively the RCFD generator shall yield the output signal with delays specified in Table I and Table II respectively. All delays shall be measured at the 50 percent points.

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TABLE I. - INPUT AND OUTPUT CHARACTERISTICS

REQUIREMENT	CHARACTERISTICS	TERMINALS
Input power	5.0 ±0.3 Vdc regulated; ac ripple less than 50 mV peak-to-peak, capacity 300 mA.	P1-30, P1-31, & P1-32 (+5 V) P1-24, P1-25, & P1-26 (5 V RET)
	25 ±0.5 Vdc regulated; ac ripple less than 50 mV peak-to-peak; capacity 100 mA.	P1-55 & P1-56 (+25 V) P1-53 & P1-54 (25 V RET)
Input signal	20 kHz positive pulse of not less than 3 Vdc amplitude, 0.4 μs wide.	P1-51
Input logic levels	See Table II.	
Output signal	20 kHz positive pulse width amplitude between 2 to 4 Vdc. 0.4 to 1.5 μs wide at 50% points; and rise time between 10% and 90% points is 60 ns max. See Table II for delay measurements.	P1-11 & P1-41

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TABLE II. - SIGNAL DELAY LIMITS

STEP	LOGIC 1 INPUT TERMINALS	SWITCH	OUTPUT TERMINALS	TEST POINT	DELAY LIMITS (ns) NLT	NMT	NOTES
1	P1-48	1	P1-41	1	30.0	75.0	1, 3
2	P1-45	2	P1-41	1	7.1	9.7	2, 3
3	P1-39	3	P1-41	1	14.0	19.0	2, 3
4	P1-35	4	P1-41	1	21.0	29.0	2, 3
5	P1-47	5	P1-41	1	28.0	28.0	2, 3
6	P1-43	6	P1-41	1	35.0	48.0	2, 3
7	P1-37	7	P1-41	1	43.0	58.0	2, 3
8	P1-23	8	P1-41	1	50.0	67.0	2, 3
9	P1-21	9	P1-11	2	57.0	77.0	2, 3
10	P1-15	10	P1-11	2	64.0	87.0	2, 3
11	P1-9	11	P1-11	2	71.0	96.0	2, 3
12	P1-5	12	P1-11	2	78.0	105.0	2, 3
13	P1-17	13	P1-11	2	85.0	115.0	2, 3
14	P1-13	14	P1-11	2	92.0	125.0	2, 3
15	P1-7	15	P1-11	2	99.0	134.0	2, 3
16	P1-3	16	P1-11	2	105.0	143.0	2, 3

NOTES: 1. Delay is from leading edge of input signal T1 specified in Table I to leading edge of output signal specified in Table I.

2. Delay is from leading edge of output signal observed in step 1 and step being presently observed.

3. Switch as specified for application of logic 1 shall be in position 1 and all others in position 2 for logic 0.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification of the circuit card assembly.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

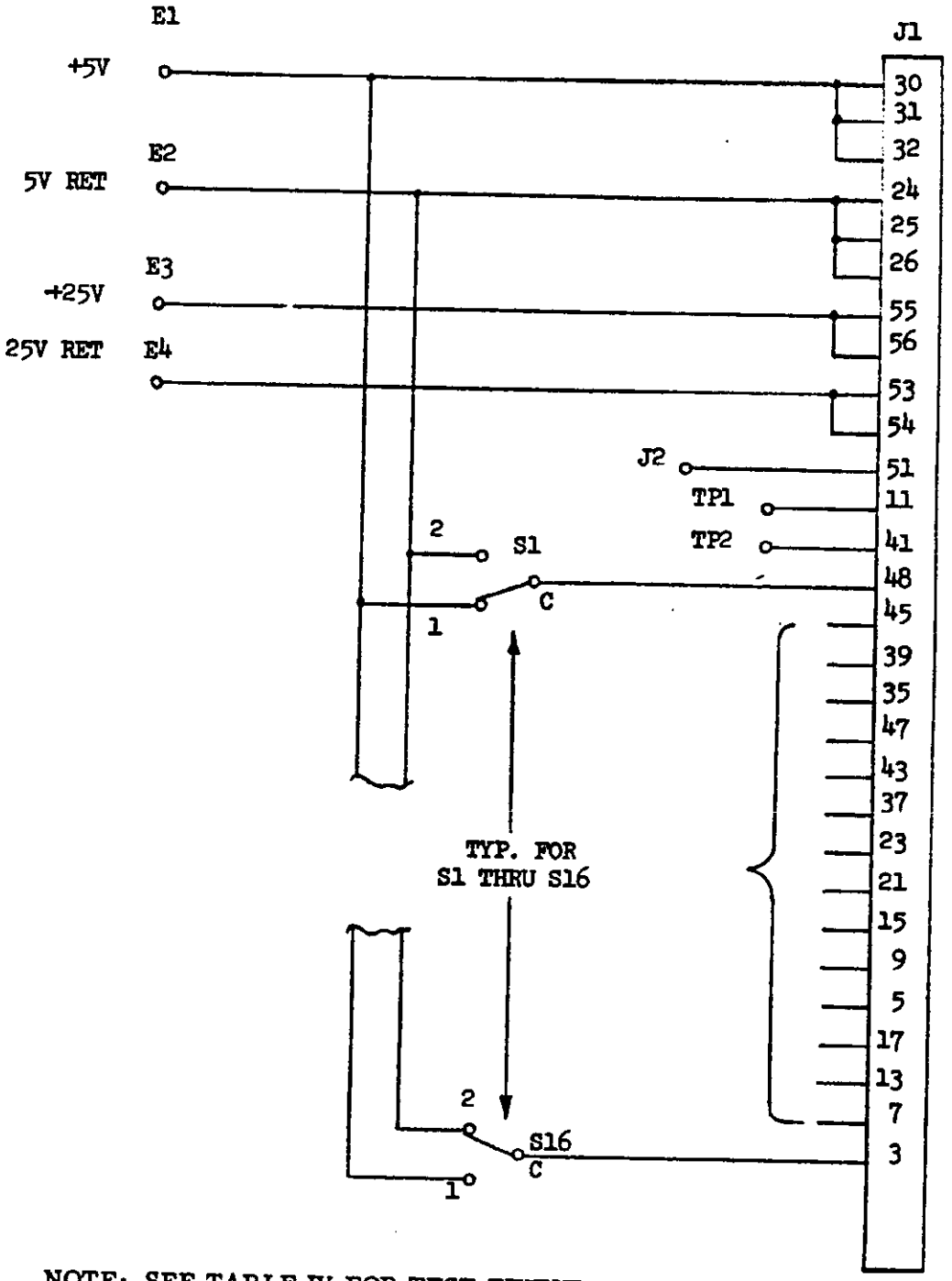
4.5.3 Test equipment. - In conjunction with 4.5 the test equipment listed in Table III shall be utilized in the performance of the applicable tests as specified in 4.6.3.

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TABLE III. - TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	Power supply	5.0 ±0.3 Vdc regulated; ac ripples less than 50 mV peak-to-peak; capacity 300 mA.
2.	Power supply	25 ±0.5 Vdc regulated; ac ripple less than 50 mV peak-to-peak; capacity 100 mA.
3.	Pulse generator	20 kHz positive pulse, amplitude 3 Vdc peak-to-peak, into 50-ohm load.
4.	Test fixture	See Figure 1. See Table IV for test fixture parts list.
5.	Oscilloscope	Dual channel; rise time of scope plus measuring probe 10 ns or less, sweep period 10 ns or less, sweep period ns per division or greater, input resistance greater than 8 megohms, input capacitance less than 25 pF.

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NOTE: SEE TABLE IV FOR TEST FIXTURE PARTS LIST

FIGURE 1. TEST FIXTURE SETUP.

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TABLE IV. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1	Binding post, H.H. Smith 257-R or equivalent.
E2	Same as E1.
E3	Binding post, H.H. Smith 257-B or equivalent.
E4	Same as E3.
J1	Connector; any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp. Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2	Connector, receptacle, UG-25B/U or equivalent.
S1 thru S16	Switch SPDT.
TP1	Tip jack, H.H. Smith 1501 or equivalent.
TP2	Same as TP1.

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4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2. Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - The selected RCFD generator shall be tested as follows:

4.6.3.1 Test setup. - Adjust power supplies and pulse generator, items 1 through 3, to deliver outputs described as input power and input signals in Table I. Connect test equipment, items 1 through 5, as shown in Figure 2. Insert RCFD generator into J1 in test fixture item 4.

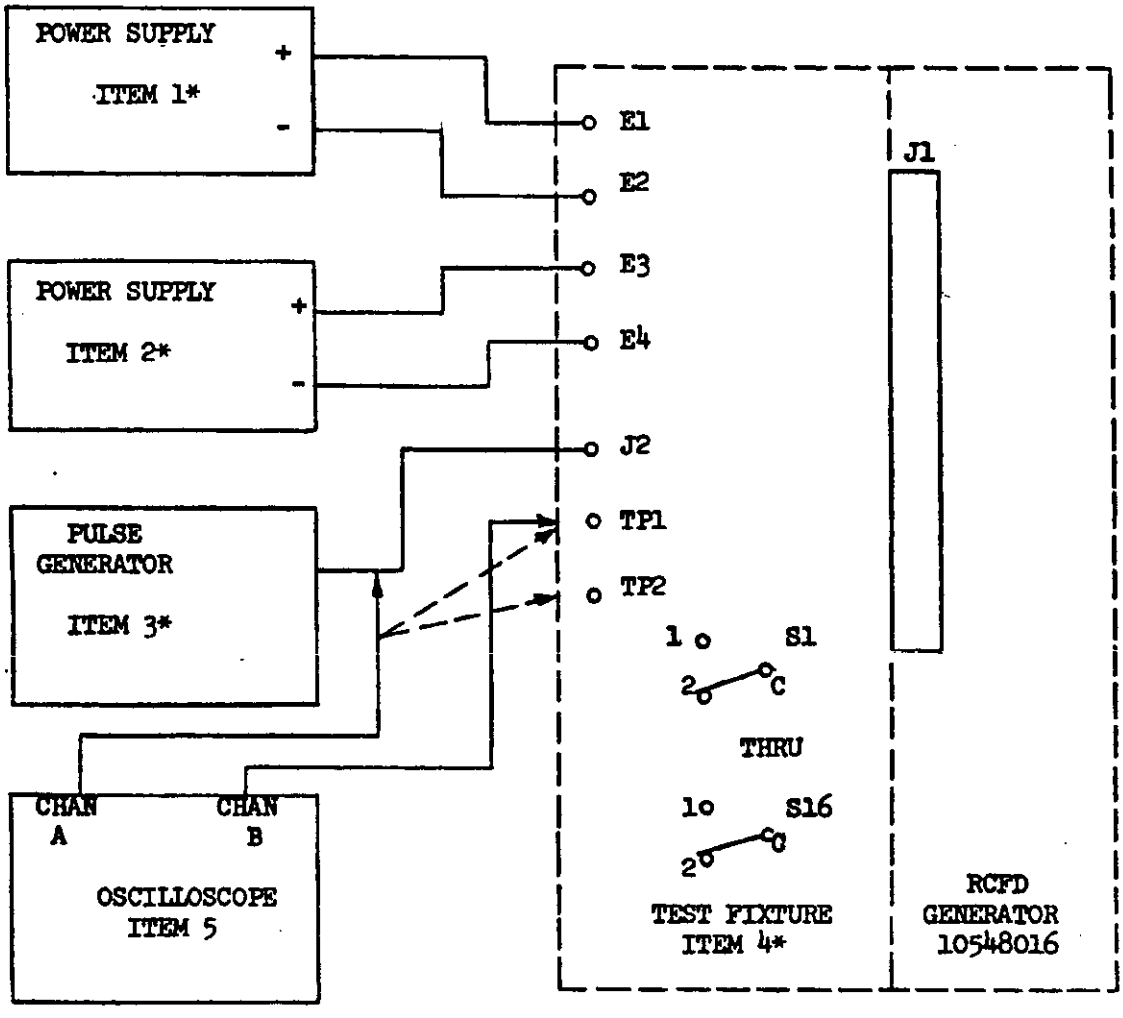
4.6.3.2 Initial delay. - Using channels A and B of oscilloscope, item 5, to measure the input and output signal, the output pulse shall be as shown in Table I with a delay as specified in Step 1 of Table II.

4.6.3.3 Gate decoder delays. - Using channel A of oscilloscope, the output pulse shall be as shown in Table I with the delays as specified in Table II for each Step 2 through 16.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548016. The level of protection shall be as specified in the procurement document.

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*REFER TO 4.5.2.3

FIGURE 2. RCFD GENERATOR TEST SETUP.

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6. NOTES

6.1 Intended use. - The RCFD generator is used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The RCFD generator provides vernier delays of ±1.25 meters of the E/L course trigger produced in the range counter assembly.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A242

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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